

Name: _____ Date: _____

Period: _____

Conic Sections Check

1. Write the standard form equation of an ellipse centered at $(3, -1)$ with horizontal radius 5 and vertical radius 2.

$$\frac{(x-3)^2}{25} + \frac{(y+1)^2}{4} = 1$$

2. For the equation $\frac{(y+2)^2}{16} - \frac{(x-1)^2}{9} = 1$:

(a) What type of conic is this? **Hyperbola (the y -term is positive and the two terms are subtracted)**

(b) What is the center? **$(1, -2)$**

(c) Write both asymptote equations.

$$y + 2 = \frac{4}{3}(x - 1) \quad \text{and} \quad y + 2 = -\frac{4}{3}(x - 1)$$

Teacher Note

Problem 1: accept any algebraically equivalent form. The most common error is writing 25 and 4 without squaring (i.e., $a = 5 \Rightarrow a^2 = 25$) — verify students wrote a^2 in the denominator, not a .

Problem 2: $b^2 = 16 \Rightarrow b = 4$ (from the positive y term) and $a^2 = 9 \Rightarrow a = 3$ (from the x term). Asymptote slope is $b/a = 4/3$. The hyperbola opens up and down because the y -term is positive.